MH1200 Linear Algebra I.

Solutions to Problem set #4.

Problem 1:

Let $\mathbf{A} = (a_{ij})_{3\times 4}$, where $a_{ij} = 2i - 3j$, $\mathbf{B} = \mathbf{I}_4$, $\mathbf{C} = \mathbf{0}_{3\times 3}$,

$$\mathbf{D} = (d_{ij})_{4\times 3} \text{ where } d_{ij} = \begin{cases} -1 & \text{if } i+j \text{ is even} \\ 1 & \text{if } i+j \text{ is odd.} \end{cases}$$

$$\mathbf{E} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{pmatrix}, \quad \mathbf{F} = \begin{pmatrix} 5 & -1 \\ 9 & 1 \\ 2 & 0 \end{pmatrix}, \quad \text{and} \quad \mathbf{G} = \begin{pmatrix} 1 \\ -1 \\ 3 \\ 2 \end{pmatrix}.$$

Evaluate the following, whenever possible.

(a) $\mathbf{AD}$, (b) $\mathbf{DA} - 3\mathbf{B}$, (c) $\mathbf{D}^2$, (d) $\mathbf{E}^2 + \mathbf{C}^3$,
(e) $\mathbf{DE} + 2\mathbf{D}$, (f) $\mathbf{EA}$, (g) $\mathbf{DB}$, (h) $\mathbf{CF}$,
(i) $\mathbf{AG}$, (j) $\mathbf{FE}$, (k) $\mathbf{EF}$, (l) $\mathbf{CA}$,
(m) $\mathbf{E} - \mathbf{E}^T$, (n) $\mathbf{F} - \mathbf{F}^T$, (o) $\mathbf{GG}^T$, (p) $\mathbf{G}^T\mathbf{G}$.

Solutions

$$\mathbf{A} = \begin{pmatrix} -1 & -4 & -7 & -10 \\ 1 & -2 & -5 & -8 \\ 3 & 0 & -3 & -6 \end{pmatrix} \quad , \quad \mathbf{D} = \begin{pmatrix} -1 & 1 & -1 \\ 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{pmatrix}$$

$(a)\ \ \mathbf{AD} = \begin{pmatrix} -6 & 6 & -6 \\ -6 & 6 & -6 \\ -6 & 6 & -6 \end{pmatrix}$ $\quad (b)\ \ \mathbf{DA} - 3\mathbf{B} = \begin{pmatrix} -4 & 2 & 5 & 8 \\ 1 & -5 & -5 & -8 \\ -1 & 2 & 2 & 8 \\ 1 & -2 & -5 & -11 \end{pmatrix}$ $\quad (c)\ \ \mathbf{D}^2$ not possible

$(d)\ \ \mathbf{E}^2 + \mathbf{C}^3 = \begin{pmatrix} 1 & 3 & 6 \\ 0 & 4 & 10 \\ 0 & 0 & 9 \end{pmatrix}$ $\quad (e)\ \ \mathbf{DE} + 2\mathbf{D} = \begin{pmatrix} -3 & 3 & -4 \\ 3 & -3 & 4 \\ -3 & 3 & -4 \\ 3 & -3 & 4 \end{pmatrix}$

$(f)\ \ \mathbf{EA} = \begin{pmatrix} 3 & -6 & -15 & -24 \\ 8 & -4 & -16 & -28 \\ 9 & 0 & -9 & -18 \end{pmatrix}$ $\quad (g)\ \ \mathbf{DB}$ not possible $\quad (h)\ \ \mathbf{CF} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$

$(i)\ \ \mathbf{AG} = \begin{pmatrix} -38 \\ -28 \\ -18 \end{pmatrix}$ $\quad (j)\ \ \mathbf{FE}$ not possible $\quad (k)\ \ \mathbf{EF} = \begin{pmatrix} 16 & 0 \\ 22 & 2 \\ 6 & 0 \end{pmatrix}$ $\quad (l)\ \ \mathbf{CA} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

$(m)\ \ \mathbf{E} - \mathbf{E}^T = \begin{pmatrix} 0 & 1 & 1 \\ -1 & 0 & 2 \\ -1 & -2 & 0 \end{pmatrix}$ $\quad (n)\ \ \mathbf{F} - \mathbf{F}^T$ not possible

$(o)\ \ \mathbf{GG}^T = \begin{pmatrix} 1 & -1 & 3 & 2 \\ -1 & 1 & -3 & -2 \\ 3 & -3 & 9 & 6 \\ 2 & -2 & 6 & 4 \end{pmatrix}$ $\quad (p)\ \ \mathbf{G}^T\mathbf{G} = \begin{pmatrix} 15 \end{pmatrix}$

$\square$

Problem 2:

Let $\mathbf{A} = (a_{ij})$ be an $m \times n$ matrix and $\mathbf{B} = (b_{ij})$ an $n \times m$ matrix, with $m, n \geq 5$.

1. Each of the following sums represents an entry of either $\mathbf{AB}$ or $\mathbf{BA}$. Determine which matrix product is involved and which entry of that product is represented in each case:

$$(i)\ \sum_{k=1}^{n} a_{3k} b_{k4} \qquad (ii)\ \sum_{p=1}^{n} a_{4p} b_{p1} \qquad (iii)\ \sum_{q=1}^{m} a_{q2} b_{3q} \qquad (vi)\ \sum_{x=1}^{m} b_{2x} a_{x5}$$

2. Use the symbol $\sum$ to express the following entries symbolically.

   (a) In $\mathbf{AB}$, the entry in the 3rd row and 2nd column.

   (b) In $\mathbf{BA}$, the entry in the 4th row and 1st column.

Solution to 1

(i) $(3,4)$ entry in AB, (ii) $(4,1)$ entry in AB, (iii) $(3,2)$ entry in BA, (iv) $(2,5)$ entry in BA.

Solution to 2

$$(a)\ \sum_{k=1}^{n} a_{3k} b_{k2} \quad , \qquad (b)\ \sum_{k=1}^{m} b_{4k} a_{k1}\ .$$

$\square$

Problem 3:

Using summation notation, prove that matrix multiplication is distributive. That is:

$$(\mathbf{A}+\mathbf{B})\,\mathbf{C}=\mathbf{AC}+\mathbf{BC}$$

and

$$\mathbf{A}\,(\mathbf{B}+\mathbf{C})=\mathbf{AB}+\mathbf{AC}.$$

(Maybe just bother proving the first equation. The second is exactly the same.)

Solution

This proof will be totally mechanical (by which I mean no clever ideas are needed).

First we'll introduce some notation for the sizes and entries of these matrices so we can discuss them in the proof. Assume:

$$\mathbf{A}=[a_{ij}]_{p\times m}\ ,\ \mathbf{B}=[b_{ij}]_{p\times m}\ ,\ \mathbf{C}=[c_{ij}]_{m\times n}\,.$$

Note that all the operations in the equation are well-defined (because $\mathbf{A}$ and $\mathbf{B}$ are the same shape and the number of columns of $\mathbf{A}$ equals the number of rows of $\mathbf{C}$). Also notice both sides of the equation will be the same shape: $p\times n$ matrices.

We'll just focus on a particular entry and calculate it and check the answer for the two sides of the equation is equal.

So consider the (k,l)-entry where k and l are such that $1\le k\le p$ and $1\le l\le n$.

The (k,l)-entry of the LHS is:

$$
\begin{aligned}
&= \textstyle\sum_{z=1}^{m} \left(\text{The } (k,z)\text{-entry of } (\mathbf{A}+\mathbf{B})\right) c_{zl} && \text{(Definition of matrix multiplication.)}\\
&= \textstyle\sum_{z=1}^{m} \left(a_{kz}+b_{kz}\right) c_{zl} && \text{(Definition of matrix addition.)}\\
&= \textstyle\sum_{z=1}^{m} a_{kz}c_{zl}+\sum_{z=1}^{m} b_{kz}c_{zl} && \text{(A calculation in } \mathbb{R}.\text{)}\\
&= (\text{The } (k,l)\text{-entry of } \mathbf{AC}) + (\text{The } (k,l)\text{-entry of } \mathbf{BC}) && \text{(Definition of matrix multiplication.)}\\
&= \text{The } (k,l)\text{-entry of } \mathbf{AC}+\mathbf{BC} && \text{(Definition of matrix addition.)}
\end{aligned}
$$

This is the (k,l)-entry of the RHS, so the proof is concluded.

$\square$

Problem 4:

In this problem we'll check that it is possible to *change the order of summation symbols* in expressions. This property is very useful when we manipulate expressions that appears during matrix calculations. Let $f(x, y)$ be some function of two variables. Check by explicitly expanding the sums on the two sides of the equation that

$$\sum_{m=1}^{5}\sum_{n=1}^{3} f(m,n) = \sum_{n=1}^{3}\sum_{m=1}^{5} f(m,n).$$

Note that the precise interpretation of a double summation like $\sum_{n=1}^{3}\sum_{m=1}^{5} f(m,n)$ is:

$$\sum_{n=1}^{3}\left(\sum_{m=1}^{5} f(m,n)\right) = \sum_{m=1}^{5} f(m,1) + \sum_{m=1}^{5} f(m,2) + \sum_{m=1}^{5} f(m,3).$$

Solution

The problem is just asking us to check a specific case.

A nice visual way to see this is just to write out all the contributing terms in a table:

$$\begin{matrix} f(1,1) & f(1,2) & f(1,3) \\ f(2,1) & f(2,2) & f(2,3) \\ f(3,1) & f(3,2) & f(3,3) \\ f(4,1) & f(4,2) & f(4,3) \\ f(5,1) & f(5,2) & f(5,3) \end{matrix}$$

We are going to add up all the entries of this table in two different ways.

The first way we'll do this is to first sum the rows, and then add up the totals for the rows. Observe that the total for the mth-row is

$$\sum_{n=1}^{3} f(m,n).$$

So the total sum we get for the table is:

$$\sum_{n=1}^{3} f(1,n) + \sum_{n=1}^{3} f(2,n) + \sum_{n=1}^{3} f(3,n) + \sum_{n=1}^{3} f(4,n) + \sum_{n=1}^{3} f(5,n).$$

This equals

$$\sum_{m=1}^{5}\sum_{n=1}^{3} f(m,n),$$

which is the left-hand-side of the equation we are checking.

On the other hand, you can check that if we first sum up the columns, and then sum up those totals, we get

$$\sum_{m=1}^{5} f(m,1) + \sum_{m=1}^{5} f(m,2) + \sum_{m=1}^{5} f(m,3) = \sum_{n=1}^{3}\sum_{m=1}^{5} f(m,n),$$

which is the right-hand-side of the equation we are checking.

$\square$

Problem 5:

Recall that the trace $\mathrm{tr}(\mathbf{A})$ of a square matrix $\mathbf{A}$ is defined to be the sum of its diagonal entries.

1. Show that the following basic properties of trace hold for any square matrices $\mathbf{A}$ and $\mathbf{B}$ of the same order, and any scalar $r \in \mathbb{R}$:

   (a) $\mathrm{tr}(\mathbf{A}+\mathbf{B}) = \mathrm{tr}(\mathbf{A}) + \mathrm{tr}(\mathbf{B})$

   (b) $\mathrm{tr}(r\mathbf{A}) = r\,\mathrm{tr}(\mathbf{A})$

   (c) $\mathrm{tr}(\mathbf{AB}) = \mathrm{tr}(\mathbf{BA})$

2. Exploit the properties of trace you proved in part 1 to prove the following fact: There are no square matrices $\mathbf{A}$ and $\mathbf{B}$ of the same order that satisfy

$$\mathbf{AB} - \mathbf{BA} = \mathbf{I}.$$

   Here $\mathbf{I}$ denotes the identity matrix of the same order as $\mathbf{A}$ and $\mathbf{B}$.

Solution to 1.

Let n denote the order of the square matrices $\mathbf{A}$ and $\mathbf{B}$ (i.e. they are $n \times n$ matrices). With the notations $\mathbf{A} = (a_{ij})$ and $\mathbf{B} = (b_{ij})$ we have $\mathbf{A}+\mathbf{B} = (a_{ij}+b_{ij})$ and $r\mathbf{A} = (ra_{ij})$. It follows that

$$\mathrm{tr}(\mathbf{A}+\mathbf{B}) = \sum_{i=1}^{n}(a_{ii}+b_{ii}) = \sum_{i=1}^{n} a_{ii} + \sum_{i=1}^{n} b_{ii} = \mathrm{tr}(\mathbf{A}) + \mathrm{tr}(\mathbf{B})$$

and

$$\mathrm{tr}(r\mathbf{A}) = \sum_{i=1}^{n} ra_{ii} = r\sum_{i=1}^{n} a_{ii} = r\,\mathrm{tr}(\mathbf{A})$$

which shows (a) and (b). To show (c), set $\mathbf{C} = \mathbf{AB}$, then $\mathbf{C} = (c_{ij})$ with $c_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj}$ and we get

$$\mathrm{tr}(\mathbf{AB}) = \mathrm{tr}(\mathbf{C}) = \sum_{i=1}^{n} c_{ii} = \sum_{i=1}^{n}\sum_{k=1}^{n} a_{ik}b_{ki}\,.$$

Similarly, set $\mathbf{D} = \mathbf{BA}$, then $\mathbf{D} = (d_{ij})$ with $d_{ij} = \sum_{m=1}^{n} b_{im}a_{mj}$ and we get

$$\mathrm{tr}(\mathbf{BA}) = \mathrm{tr}(\mathbf{D}) = \sum_{j=1}^{n} d_{jj} = \sum_{j=1}^{n}\sum_{m=1}^{n} b_{jm}a_{mj}\,.$$

The last expression can be rewritten as

$$\sum_{j=1}^{n}\sum_{m=1}^{n} b_{jm}a_{mj} = \sum_{j=1}^{n}\sum_{m=1}^{n} a_{mj}b_{jm} = \sum_{m=1}^{n}\sum_{j=1}^{n} a_{mj}b_{jm} \overset{\substack{m\to i\\ j\to k}}{=} \sum_{i=1}^{n}\sum_{k=1}^{n} a_{ik}b_{ki}$$

This is precisely the expression for $\mathrm{tr}(\mathbf{AB})$ found above. Thus we have shown $\mathrm{tr}(\mathbf{BA}) = \mathrm{tr}(\mathbf{AB})$.

Solution to 2.

If $\mathbf{A}$ and $\mathbf{B}$ satisfy this relation, then, taking the trace of each side, we get

$$\mathrm{tr}(\mathbf{AB} - \mathbf{BA}) = \mathrm{tr}(\mathbf{I}).$$

We now show that this equality cannot be true, which implies that square matrices satisfying the relation above do not exist. On the one hand, the $n \times n$ identity matrix $\mathbf{I}$ has n entries in its diagonal, and they are all 1, so $\mathrm{tr}(\mathbf{I}) = n$. On the other hand, using the properties of the trace shown in part 1, we find

$$\mathrm{tr}(\mathbf{AB} - \mathbf{BA}) = \mathrm{tr}(\mathbf{AB}) - \mathrm{tr}(\mathbf{BA}) = \mathrm{tr}(\mathbf{AB}) - \mathrm{tr}(\mathbf{AB}) = 0 \neq n$$

So the equality above cannot hold for any square matrices $\mathbf{A}$ and $\mathbf{B}$.

$\square$

Problem 6:

Let $\mathbf{I}_n$ denote the $n \times n$-identity matrix. In other words this is the matrix whose (i, j)-entry is

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

Prove that for every $m \times n$ matrix $\mathbf{A}$

$$\mathbf{I}_m \mathbf{A} = \mathbf{A} \mathbf{I}_n = \mathbf{A}.$$

Solution

The symbol δ_{ij} introduced here is a common workhorse of mathematics called the Kronecker delta. It is typically used to collapse sums to one of their summands in the following way:

$$\sum_{k=1}^{m} \delta_{jk} F_k = F_j.$$

This equation can be easily seen if your write it out the sum

$$\delta_{j1} F_1 + \delta_{j2} F_2 + \ldots + \delta_{jj} F_j + \ldots + \delta_{jm} F_m.$$

In all these terms the δ_{jk} factor is zero, except the (jj) term where it is 1. Thus this sum equals F_j.

Turning to this problem - By the formula for matrix multiplication the (i, j)-entry of the product $\mathbf{I}_m \mathbf{A}$ equals

$$\sum_{k=1}^{m} \delta_{ik} \mathbf{A}_{kj} = \mathbf{A}_{ij}.$$

Because all the corresponding entries on the LHS and the RHS of the matrix equation are equal, the matrix equation is proved.

□

Problem 7: (Appeared on 2022 final exam.)

Consider the following matrix equation depending on 4 real variables a, b, c and d. Determine the set of all possible solutions of this equation. (In other words, determine the set of all $(a, b, c, d) \in \mathbb{R}^4$ such that the equation is true when those values are substituted into the variables.)

$$a\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} + b\begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix} = c\begin{bmatrix} 3 & 1 \\ -1 & 3 \end{bmatrix} + d\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Show in full detail all the steps needed to solve these equations without a calculator, although you may check your answer with a calculator if you wish.

Solution

If we perform matrix operations to express the two sides of the equation as a single matrix we get:

$$\begin{bmatrix} a & -a+2b \\ -a+b & a \end{bmatrix} = \begin{bmatrix} 3c+d & c+d \\ -c & 3c+d \end{bmatrix}.$$

Equating corresponding entries on the two sides of the equation we get a linear system of 4 equations in the variables.

$$\begin{aligned} a - 3c - d &= 0 \\ -a + 2b - c - d &= 0 \\ -a + b + c &= 0 \\ a - 3c - d &= 0. \end{aligned}$$

Now we solve this system. The augmented matrix is:

$$\left[\begin{array}{cccc|c} 1 & 0 & -3 & -1 & 0 \\ -1 & 2 & -1 & -1 & 0 \\ -1 & 1 & 1 & 0 & 0 \\ 1 & 0 & -3 & -1 & 0 \end{array}\right].$$

Following Gaussian elimination:

$$\left[\begin{array}{cccc|c} 1 & 0 & -3 & -1 & 0 \\ 0 & 2 & -4 & -2 & 0 \\ 0 & 1 & -2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & -3 & -1 & 0 \\ 0 & 1 & -2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Now we introduce two variables: $c = s$ and $d = t$. And solve for:

$$b = 2s + t.$$

And:

$$a = 3s + t$$

The set of solutions is:

$$\{(3s + t, 2s + t, s, t); s, t \in \mathbb{R}\}.$$

□

Problem 8:

Let $\mathbf{A}$ be any diagonal matrix of order n. Show that a matrix $\mathbf{B}$ with the property that $\mathbf{AB} = \mathbf{I}$ exists if and only if $a_{ii} \neq 0$ for all i.

Solution

We first show that if all $a_{ii} \neq 0$ then $\mathbf{B}$ with $\mathbf{AB} = \mathbf{I}$ exists. Simply let $b_{ii} = 1/a_{ii}$. Then the (i, i)-th entry of $\mathbf{AB}$ is $\sum_{k=1}^{n} a_{ik}b_{ki} = a_{ii}b_{ii} = 1$. On the other hand for $i \neq j$ the $(i.j)$-th entry of $\mathbf{AB}$ is $\sum_{k=1}^{n} a_{ik}b_{kj} = a_{ii}b_{ij} + a_{ij}b_{jj} = 0$ and $\mathbf{AB} = \mathbf{I}$.

For the other direction we need to show that if there is an i with $a_{ii} = 0$ then $\mathbf{B}$ does not exist. So assume that $\mathbf{B}$ exists, then the (i, i)-th entry of $\mathbf{AB}$ is again $\sum_{k=1}^{n} a_{ik}b_{ki} = a_{ii}b_{ii} = 0$. So the product $\mathbf{AB} \neq \mathbf{I}$.

□

Problem 9: (Part (ii) appeared on 2018 Midterm.)

(i) Let $\mathbf{A}$ and $\mathbf{B}$ be matrices such that the matrix product $\mathbf{AB}$ is well-defined. Explain why the matrix product $\mathbf{B}^T\mathbf{A}^T$ is also well-defined and prove that

$$(\mathbf{AB})^T = \mathbf{B}^T\mathbf{A}^T.$$

(ii) Consider a square matrix $\mathbf{A}$. Show that $\mathbf{A}$ is invertible if and only if its transpose is invertible.

Solution to (i)

Let's introduce notation: assume $\mathbf{A} = [a_{ij}]_{m\times n}$ and $\mathbf{B} = [b_{ij}]_{n\times p}$. We are told that the product $\mathbf{AB}$ is well-defined which is why the number of columns of $\mathbf{A}$ (namely n) equals the number of rows of $\mathbf{B}$ (also n).

The first question is why is the product $\mathbf{B}^T\mathbf{A}^T$ well-defined? The reason is that:

$$(\#\text{ of columns of }\mathbf{B}^T) = (\#\text{ of rows of }\mathbf{B}) = (\#\text{ of columns of }\mathbf{A}) = (\#\text{ of rows of }\mathbf{A}^T).$$

Next we must prove the equation: $(\mathbf{AB})^T = \mathbf{B}^T\mathbf{A}^T$. As usual we do this by showing that every entry of the two sides of the equation are equal.

The two sides are $p \times m$ matrices. Pick an entry (i, j) with $1 \leq i \leq p$ and $1 \leq j \leq m$. Then:

$$\begin{aligned}
&\text{The } (i,j)\text{-entry of } (\mathbf{AB})^T \\
&= \text{The } (j,i)\text{-entry of } \mathbf{AB} \\
&= \sum_{z=1}^{n} a_{jz}b_{zi} \\
&= \sum_{z=1}^{n} \left(\text{The } (z,j)\text{-entry of } \mathbf{A}^T\right)\left(\text{The } (i,z)\text{-entry of } \mathbf{B}^T\right) \\
&= \sum_{z=1}^{n} \left(\text{The } (i,z)\text{-entry of } \mathbf{B}^T\right)\left(\text{The } (z,j)\text{-entry of } \mathbf{A}^T\right) \\
&= \text{The } (i,j)\text{-entry of } \mathbf{B}^T\mathbf{A}^T.
\end{aligned}$$

Solution to (ii).

We are told that there is a square matrix $\mathbf{A}$ which is invertible. What is the definition of this? It means: there exists a matrix $\mathbf{A}^{-1}$ such that

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I} \quad (\star).$$

Our problem is to prove an "if and only if" statement. This we means we have two directions to prove.

Part 1: Forward direction. Prove that $\mathbf{A}$ is invertible implies $\mathbf{A}^T$ is also invertible.

Take the transpose of equation ($\star$) using the rule we proved in Part (i) of this problem. We deduce:

$$(\mathbf{A}^{-1})^T\mathbf{A}^T = \mathbf{A}^T(\mathbf{A}^{-1})^T = \mathbf{I}^T = \mathbf{I}.$$

To make this clearer we could set $\mathbf{B} = (\mathbf{A}^{-1})^T$. Then the equations become

$$\mathbf{B}\mathbf{A}^T = \mathbf{A}^T\mathbf{B} = \mathbf{I}^T = \mathbf{I}.$$

It is clear from this that $\mathbf{B}$, namely $(\mathbf{A}^{-1})^T$, is an inverse of $\mathbf{A}^T$.

Alternatively we write: $\left(\mathbf{A}^T\right)^{-1} = (\mathbf{A}^{-1})^T$.

Part 2: Backward direction. Prove that $\mathbf{A}^T$ is invertible implies $\mathbf{A}$ is also invertible.

So in this case we assume that $\mathbf{A}^T$ is invertible.

If we apply the deductions of Part 1 to this fact we deduce that $\left(\mathbf{A}^T\right)^T$ is also invertible. But the transpose of a transpose is just the original matrix, so we conclude that $\left(\mathbf{A}^T\right)^T = \mathbf{A}$ is invertible as required.

$\square$

Problem 10:

In lectures we met the concept of a **lower triangular** matrix. This is a square matrix satisfying the property that the entries a_{ij} where $j > i$ are all zero. (In other words, the entries lying strictly above the diagonal are all zero.)

Prove that if both $\mathbf{A}$ and $\mathbf{B}$ are lower triangular matrices of the same order, then so is their product $\mathbf{AB}$.

Use transpose to immediately deduce a similar property for upper triangular matrices.

Solution

We'll start by introducing some notation so we can discuss $\mathbf{A}$ and $\mathbf{B}$. Assume that

$$\mathbf{A} = [a_{ij}]_{n\times n} \quad \text{and} \quad \mathbf{B} = [b_{ij}]_{n\times n}.$$

We know these matrices are lower-triangular. This means

$$a_{ij} = b_{ij} = 0 \ \text{ when } j > i \qquad (\star).$$

To test if the product $\mathbf{AB}$ is also lower triangular we must test one of the entries lying above the diagonal. Namely, we must calculate the (k, l)-entry of $\mathbf{AB}$ where $l > k$ (call this assumption that $l > k$ $(\star\star)$). It equals:

$$\sum_{z=1}^{n} a_{kz} b_{zl}.$$

Now by $(\star)$, for one of the terms in this sum $a_{kz}b_{zl}$ to be non-zero we require that **both** $z \leq k$ and $l \leq z$. But these two conditions together imply $l \leq k$, which contradicts $(\star\star)$.

Thus all the entries in this sum are zero.

$\square$

Problem 11: (This was the hardest problem on the 2018 final.)

A square matrix A is said to be

- an involutory matrix if $A^2 = I$,
- an idempotent if $A^2 = A$.

Show that every involutory matrix can be expressed as a difference of two idempotents.

Solution

The required decomposition is:

$$A = \left(\frac{1}{2}(I + A)\right) - \left(\frac{1}{2}(I - A)\right).$$

It is straightforward to check these two terms are both idempotents.

$\square$

Every year many students always ask me where this solution came from. In a previous year I wrote the following on the discussion board about this problem:

"Actually a very large amount of mathematics needs inspired guesswork and trial and error. The tidy proofs and explanation that come after the fact often hide the way a proof was first guessed.

If I was tackling this problem for the first time, I might ask myself: How can I build an idempotent $X^2 = X$ from an involutory matrix $A^2 = I$? I might notice that because $A^2 = I$ then any matrix I build of the form $rI + sA$ and every sum and product is also going to be of that form. Maybe I can solve for r and s? If we try guessing $X = rI + sA$ then

$$X^2 = (rI + sA)(rI + sA) = ((r^2) + (s^2))I + 2rsA.$$

Solving this for coefficients immediately gives me: $r^2 + s^2 = r$ and $2rs = s$. So $r = 1/2$. $s = \pm 1/2$."

Problem 12: (Parts (i) and (ii) appeared on a 2020 midterm.)

(i) Consider the matrix

$$\mathbf{P} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Compute $\mathbf{P}^3 - \mathbf{I}_3$. Show the steps in your working.

(ii) Let a and b be real numbers. Calculate the following quantity:

$$\operatorname{tr}\left(\begin{bmatrix} a & 0 & b \\ b & a & 0 \\ 0 & b & a \end{bmatrix}^9\right).$$

(iii) Let a and b be real numbers. Give a closed expression for the following quantity as a polynomial in a and b.

$$\operatorname{tr}\left(\begin{bmatrix} a & 0 & b \\ b & a & 0 \\ 0 & b & a \end{bmatrix}^{100}\right).$$

Solution to (i)

This is just a computation: $\mathbf{P}^3 - \mathbf{I}_3$ equals

$$\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

This equals

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Which equals:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

the 3×3 zero matrix.

Solution to (ii)

We can rewrite the matrix

$$\begin{bmatrix} a & 0 & b \\ b & a & 0 \\ 0 & b & a \end{bmatrix} = a\mathbf{I}_3 + b\mathbf{P}.$$

When we raise this matrix to the power 9 we can calculate the result using the following identity about polynomials. This is possible to calculate directly (it took me less than 5 minutes), or better still you can use the binomial theorem to write the answer straight down. The coefficients are of course binomial coefficients.

$$\begin{aligned}(x+y)^9 &= x^9 + 9x^8y + 36x^7y^2 \\ &\quad +84x^6y^3 + 126x^5y^4 + 126x^4y^5 \\ &\quad +84x^3y^6 + 36x^2y^7 + 9xy^8 + y^9.\end{aligned}$$

Using this identity we deduce that $\begin{bmatrix} a & 0 & b \\ b & a & 0 \\ 0 & b & a \end{bmatrix}^9$ equals

$$\begin{aligned}&a^9\mathbf{I}_3 + 9a^8b\mathbf{P} + 36a^7b^2\mathbf{P}^2 + 84a^6b^3\mathbf{P}^3 \\ &+126a^5b^4\mathbf{P}^4 + 126a^4b^5\mathbf{P}^5 + 84a^3b^6\mathbf{P}^6 \\ &+36a^2b^7\mathbf{P}^7 + 9ab^8\mathbf{P}^8 + b^9\mathbf{P}^9.\end{aligned}$$

(Here we have depended on the fact $\mathbf{IP} = \mathbf{P} = \mathbf{PI}$ so the expansion can be done just treating $\mathbf{I}$ and $\mathbf{P}$ like normal variables in polynomial arithmetic.)

Using the fact from Part (i) that $P^3 = I$ we can rewrite this:

$$\begin{aligned}&(a^9 + 84a^6b^3 + 84a^3b^6 + b^9)\mathbf{I}_3 \\ &+(9a^8b + 126a^5b^4 + 36a^2b^7)\mathbf{P} \\ &+(36a^7b^2 + 126a^4b^5 + 9ab^8)\mathbf{P}^2.\end{aligned}$$

Trace is a linear function. So trace of the above expression decomposes into the sum of the traces of the pieces. So to determine this expression it will be enough to know the terms $\mathrm{tr}(P^k)$ where $k = 0, 1, 2$.

It follows from the calculation in Part (i) that:

- $\mathrm{tr}(P^0) = 3$
- $\mathrm{tr}(P^1) = 0$
- $\mathrm{tr}(P^2) = 0$

Thus:

$$\mathrm{tr}\left(\begin{bmatrix} a & 0 & b \\ b & a & 0 \\ 0 & b & a \end{bmatrix}^9\right) = 3(a^9 + 84a^6b^3 + 84a^3b^6 + b^9). \qquad \square$$

Solution to (iii)

In this part we have no choice but to use binomial coefficients. We will calculate:

$$\operatorname{tr}\left(\begin{bmatrix} a & 0 & b \\ b & a & 0 \\ 0 & b & a \end{bmatrix}^{100}\right) = \operatorname{tr}\left((a\mathbf{I} + b\mathbf{P})^{100}\right).$$

Expanding part of this with the binomial theorem

$$(a\mathbf{I} + b\mathbf{P})^{100} = (a\mathbf{I} + b\mathbf{P})^{100} = \sum_{n=0}^{100} \binom{100}{n} a^{100-n} b^n \mathbf{I}^{100-n}\mathbf{P}^n$$

and using linearity of the trace this becomes:

$$\operatorname{tr}\left(\sum_{n=0}^{100} \binom{100}{n} a^{100-n} b^n \mathbf{I}^{100-n}\mathbf{P}^n\right) = \sum_{n=0}^{100} \binom{100}{n} a^{100-n} b^n \operatorname{tr}\left(\mathbf{P}^n\right).$$

Only the terms in this sum where n is divisible by 3 have non-zero trace, and in this case the trace is 3. We set $n = 3m$ and sum m from 0 to 33 to enumerate the n's such that $0 \leq n \leq 100$ and n is divisible by 3.

Thus this sum equals:

$$3\sum_{m=0}^{33} \binom{100}{3m} a^{100-3m} b^{3m}.$$

$\square$

Solution to Problem 13 on the Fibonacci sequence:

This is the problem on the Fibonacci sequence $F_0 = 0$, $F_1 = 1$, $F_k = F_{k-1} + F_{k-2}$ for $k \geq 2$. To begin we will prove that for all natural numbers $k \geq 0$

$$\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}^k \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} F_k \\ F_{k+1} \end{bmatrix} \quad (\star).$$

The most natural approach here is to use mathematical induction to prove this. The base case will be $k = 0$. This case is the following fact which is trivially true by definition:

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} F_0 \\ F_1 \end{bmatrix}.$$

Now the induction step. We assume that the equation $(\star)$ is true for some k. Then:

$$\begin{aligned}
\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}^{k+1} \begin{bmatrix} 0 \\ 1 \end{bmatrix} &= \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}^k \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\
&= \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} F_k \\ F_{k+1} \end{bmatrix} \quad \text{(By the induction assumption } (\star)) \\
&= \begin{bmatrix} F_{k+1} \\ F_k + F_{k+1} \end{bmatrix} \\
&= \begin{bmatrix} F_{k+1} \\ F_{k+2} \end{bmatrix}.
\end{aligned}$$

The next step is just a direct calculation that

$$\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} = P \begin{bmatrix} \phi_+ & 0 \\ 0 & \phi_- \end{bmatrix} P^{-1}$$

where

$$\phi_{\pm} = \frac{1 \pm \sqrt{5}}{2}$$

and

$$P = \begin{bmatrix} 1 & 1 \\ \phi_+ & \phi_- \end{bmatrix}.$$

To do this we calculate the right hand side of this equation as a straightforward matrix product. For this we will need the inverse matrix P^{-1}. The formula for the inverse of a 2x2 matrix needs the determinant which is:

$$\det\left(\begin{bmatrix} 1 & 1 \\ \phi_+ & \phi_- \end{bmatrix}\right) = \phi_- - \phi_+ = -\sqrt{5}.$$

The inverse is:

$$P^{-1} = -\frac{1}{\sqrt{5}} \begin{bmatrix} \phi_- & -1 \\ -\phi_+ & 1 \end{bmatrix}$$

Now calculate:

$$
\begin{aligned}
& -\frac{1}{\sqrt{5}}\begin{bmatrix} 1 & 1 \\ \phi_+ & \phi_- \end{bmatrix}\begin{bmatrix} \phi_+ & 0 \\ 0 & \phi_- \end{bmatrix}\begin{bmatrix} \phi_- & -1 \\ -\phi_+ & 1 \end{bmatrix} \\
= & -\frac{1}{\sqrt{5}}\begin{bmatrix} \phi_+ & \phi_- \\ \phi_+^2 & \phi_-^2 \end{bmatrix}\begin{bmatrix} \phi_- & -1 \\ -\phi_+ & 1 \end{bmatrix} \\
= & -\frac{1}{\sqrt{5}}\begin{bmatrix} 0 & -\phi_+ + \phi_- \\ \phi_+\phi_-(\phi_+ - \phi_-) & -\phi_+^2 + \phi_-^2 \end{bmatrix} \\
= & \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}.
\end{aligned}
$$

With this expression in hand we can write down an exact formula for F_k:

$$
\begin{bmatrix} F_k \\ F_{k+1} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}^k \begin{bmatrix} 0 \\ 1 \end{bmatrix} = P\begin{bmatrix} \phi_+ & 0 \\ 0 & \phi_- \end{bmatrix}P^{-1}P\begin{bmatrix} \phi_+ & 0 \\ 0 & \phi_- \end{bmatrix}P^{-1}\dots P\begin{bmatrix} \phi_+ & 0 \\ 0 & \phi_- \end{bmatrix}P^{-1}\begin{bmatrix} 0 \\ 1 \end{bmatrix}
$$

When we cancel all the pairs $P^{-1}P$ that appear in this expression we get:

$$
\begin{bmatrix} F_k \\ F_{k+1} \end{bmatrix} = P\begin{bmatrix} \phi_+ & 0 \\ 0 & \phi_- \end{bmatrix}^k P^{-1}\begin{bmatrix} 0 \\ 1 \end{bmatrix}
$$

Thus:

$$
\begin{bmatrix} F_k \\ F_{k+1} \end{bmatrix} = -\frac{1}{\sqrt{5}}\begin{bmatrix} 1 & 1 \\ \phi_+ & \phi_- \end{bmatrix}\begin{bmatrix} \phi_+^k & 0 \\ 0 & \phi_-^k \end{bmatrix}\begin{bmatrix} \phi_- & -1 \\ -\phi_+ & 1 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix}
$$

Performing this multiplication gives us:

$$
\begin{bmatrix} F_k \\ F_{k+1} \end{bmatrix} = -\frac{1}{\sqrt{5}}\begin{bmatrix} 1 & 1 \\ \phi_+ & \phi_- \end{bmatrix}\begin{bmatrix} -\phi_+^k \\ \phi_-^k \end{bmatrix}.
$$

In conclusion we obtain an exact formula for F_k:

$$
F_k = \frac{1}{\sqrt{5}}\left(\phi_+^k - \phi_-^k\right).
$$

Now note that $|\phi_-| < 1$ so

$$
\lim_{k\to\infty}\left(F_k - \frac{1}{\sqrt{5}}\phi_+^k\right) = -\lim_{k\to\infty}\frac{1}{\sqrt{5}}\phi_-^k = 0.
$$

$\square$